# ($<\mathbb{N}$)-SCRAMBLED CANTOR SETS

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Abstract. We provide a homeomorphism of Cantor space that has a single uncountable set which is $d$-scrambled for all $d \geq 3$, but which does not have a 3-scrambled Cantor set.

## Introduction

Given a metric space $X$, $f\colon X \to X$, and $d \geq 2$, a sequence $x \in X^d$ is *proximal* if $\liminf_{n\to\infty} \max_{i<j<d} d_X(f^n(x(i)), f^n(x(j))) = 0$, and *separated* if $\limsup_{n\to\infty} \min_{i<j<d} d_X(f^n(x(i)), f^n(x(j))) > 0$. A set $Y \subseteq X$ is *$d$-scrambled* if every injective sequence in $Y^d$ is Li–Yorke. We say that $Y$ is ($<\mathbb{N}$)-*scrambled* if it is $d$-scrambled for all $d \geq 2$.

In [GGM21], it was shown that uncountable 2-scrambled sets yield 2-scrambled Cantor sets in Polish dynamical systems. Here we show:

**Theorem 1** ($\mathtt{ZFC}$)**.** *There is a homeomorphism of $2^{\mathbb{N}}$ with an uncountable ($<\mathbb{N}$)-scrambled set but no 3-scrambled Cantor set.*

A *graph* on $X$ is an irreflexive symmetric set $G \subseteq X \times X$. A set $Y \subseteq X$ is a *$G$-clique* if its distinct points are $G$-related, and *$G$-independent* if its points are not $G$-related. Let $\mathrm{CLQ}_{\mathrm{G}}^{\geq 3}$ and $\mathrm{IND}_{\mathrm{G}}^{\geq 3}$ denote the sets of finite sequences of length at least three whose images have these properties. Let $\mathrm{LY_f}$ denote the set of Li-Yorke sequences. A *homomorphism* from a relation $R$ on $X$ to a relation $S$ on $Y$ is an injection $\pi\colon X \to Y$ whose compositions with sequences in $R$ are in $S$. Such a function is a *homomorphism* from $(R, R')$ to $(S, S')$ if it also a homomorphism from $R'$ to $S'$. The *saturation* of a set $Y \subseteq X$ under a bijection $T\colon X \to X$ is $[Y]_T = \bigcup_{n\in\mathbb{Z}} T^n(Y)$. A subset of a topological space is $F_\sigma$ if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is the following:

**Theorem 2** ($\mathtt{ZF} + \mathtt{DC}$)**.** *For every $F_\sigma$ graph $G$ on $2^{\mathbb{N}}$, there is a continuous injective homomorphism $\pi\colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ from $(\mathrm{CLQ}_{\mathrm{G}}^{\geq 3}, \mathrm{IND}_{\mathrm{G}}^{\geq 3})$ to $(\mathrm{LY_T}, \sim\mathrm{LY_T})$ such that $|\sim[\pi(2^{\mathbb{N}})]_T| = 1$.*

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In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

## 1. Embedding hypergraphs

Define $S \colon 2^{\mathbb{Z}} \to 2^{\mathbb{Z}}$ by $S(c)(i) = c(i+1)$ for all $i \in \mathbb{Z}$ and $c \in 2^{\mathbb{Z}}$. Let $\mathrm{SEP}_{\mathrm{f}}$ denote the of separated sequences. Theorem 2 follows from the usual topological characterization of Cantor space and:

**Theorem 1.1** ($\mathtt{ZF} + \mathtt{DC}$). *Suppose that $G$ is an $F_\sigma$ graph on $2^{\mathbb{N}}$. Then there are an $S$-invariant perfect set $F \subseteq 2^{\mathbb{Z}}$, whose finite sequences are proximal, and a continuous injective homomorphism $\pi \colon 2^{\mathbb{N}} \to F$ from $(\mathrm{CLQ}_{\mathrm{G}}^{\geq 3}, \mathrm{IND}_{\mathrm{G}}^{\geq 3})$ to $(\mathrm{SEP}_{\mathrm{S}}, {\sim}\mathrm{SEP}_{\mathrm{S}})$ such that $F \setminus [\pi(2^{\mathbb{N}})]_T = \{0^{\mathbb{Z}}\}$.*

*Proof.* For all $s \in 2^{<\mathbb{N}}$, define $\mathcal{N}_s = \{c \in 2^{\mathbb{N}} \mid \forall i < |s| \; s(i) = c(i)\}$. For all $S \subseteq 2^{<\mathbb{N}}$, define $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$. For all $d \geq 3$, $\ell \in \mathbb{N}$, and $p \colon 2^\ell \to d$, define $\ell_p = \ell$ and $\mathcal{N}_p = \prod_{i<d} \mathcal{N}_{p^{-1}(\{i\})}$. We say that $p \colon 2^\ell \to d$ is a *near bijection* if it is surjective and there is a unique $k_p < d$ for which $|p^{-1}(\{k_p\})| > 1$. For all $k \in d \setminus \{k_p\}$, set $\{p^{-1}(k)\} = p^{-1}(\{k\})$.

Fix closed graphs $G_n$ on $2^{\mathbb{N}}$ for which $G = \bigcup_{n \in \mathbb{N}} G_n$. For all $n \in \mathbb{N}$, define $G_{m,n} = \{(c, c') \in 2^{\mathbb{N}} \times 2^{\mathbb{N}} \mid G_n \cap (\mathcal{N}_{c \upharpoonright m} \times \mathcal{N}_{c' \upharpoonright m}) \neq \emptyset\}$. For all $d \geq 3$ and $m, n \in \mathbb{N}$, Let $P_{d,m,n}$ denote the of all near bijections $p \colon 2^m \to d$ such that $k_p = 0 = p(0^m)$ and $\mathcal{N}_{p^{-1}(j)} \times \mathcal{N}_{p^{-1}(k)} \subseteq G_{m,n}$ for all $0 < j < k < d$. Then there is an infinite set $I \subseteq \mathbb{N} \setminus \{0\}$ and a bijection $\phi \colon I \to \bigcup_{d \geq 3, m,n \in \mathbb{N}} \{d\} \times \{m\} \times \{n\} \times P_{d,m,n}$ such that $I \cap [i, 2i + (d-2)(\phi(i)(1) + 1)] = \{i\}$ for all $i \in I \setminus \phi^{-1}(\mathbb{N})$. Define a continuous function $\pi \colon 2^{\mathbb{N}} \to 2^{\mathbb{Z}}$ by setting $\pi(c)(i) = 1$ if and only if

(1) $i = 0$;
(2) $i \in I$, $\phi(i) \in \mathbb{N}$, and $c(\phi(i)) = 1$; or
(3) $\exists d \geq 3 \exists 0 < k < d \exists m, n \in \mathbb{N} \exists p \in P_{m,n}$
$((d, m, n, p) = \phi(i - (k-1)(n+1))$ and $p(c \upharpoonright \ell_p) = k)$.

**Lemma 1.2.** *$\pi$ is injective.*

*Proof.* Suppose that $c, c' \in 2^{\mathbb{N}}$ are distinct. Then there exists $n \in \mathbb{N}$ with $c(n) \neq c'(n)$. Fix $i \in I$ for which $\phi(i) = n$. As $\pi(c)(i) = c(n)$ and $\pi(c')(i) = c'(n)$, it follows that $\pi(c)(i) \neq \pi(c')(i)$, so $\pi(c) \neq \pi(c')$. $\square$

The *product metric* on $2^{\mathbb{Z}}$ is given by $\rho(c, c') = \sum_{i \in \mathbb{Z}} |c(i) - c'(i)|/2^{|i|}$ for all $c, c' \in 2^{\mathbb{Z}}$. Define $N_i = \{i\}$ for all $i \in \phi^{-1}(\mathbb{N})$ and $N_i = [i, i + (d-2)(\phi(i)(1) + 1)]$ for all $i \in I \setminus \phi^{-1}(\mathbb{N})$.

**Lemma 1.3.** *$\pi$ is a homomorphism from $\mathrm{CLQ}_{\mathrm{G}}^{\geq 3}$ to $\mathrm{SEP}_{\mathrm{S}}$.*

*Proof.* Suppose that $c \in (2^{\mathbb{N}})^d$ is in $\mathrm{CLQ}_{\mathrm{G}}^{\geq 3}$. Fix $n \in \mathbb{N}$ sufficiently large that $c(j) \; G_n \; c(k)$ for all $j < k < d$. By permuting the coordinates of $c$, we can assume that if $c(k) = 0^{\mathbb{N}}$ for some $k < d$, then $k = 0$. Suppose that $m \in \mathbb{N}$ is sufficiently large that $\mathrm{proj}_{2^m} \circ c$ is injective, $c(k) \upharpoonright m \neq 0^m$ for all $0 < k < d$, and $d < 2^m$. Let $p \in P_{d,m,n}$ be the near bijection extending $(\mathrm{proj}_{2^m} \circ c)^{-1}$ for which $k_p = 0$ and $\ell_p = m$. Fix $i \in I$ such that $\phi(i) = (d, m, n, p)$ and observe that $\mathrm{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j<k<d} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(d-2)(n+1)}$. □

**Lemma 1.4.** *$\pi$ is a homomorphism from $\mathrm{IND}^{\geq 3}_{\mathrm{G}}$ to $\sim\mathrm{SEP}_{\mathrm{S}}$.*

*Proof.* Suppose that $c \in (2^{\mathbb{N}})^d$ and $\pi \circ c$ is separated. Fix $\epsilon > 0$ such that $I' = \{i' \in \mathbb{N} \mid \min_{j<k<d} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$ is infinite, as well as $r \in \mathbb{N}$ such that $\sum_{|i|>r} 1/2^{|i|} < \epsilon$. By throwing out finitely many elements of $I'$, we can assume that for all $i' \in I'$, there is at most one $i \in I$ such that $[i' - r, i' + r] \cap N_i \neq \emptyset$. Our choice of $r$ then yields a unique such $i \in I$, in which case $\mathrm{proj}_{2^{N_i}} \circ \pi \circ c$ is injective. As $d \geq 3$, there exist $j < k < d$, $m, n \in \mathbb{N}$, $p \in P_{d,m,n}$, and $\sigma \in \mathrm{Sym}(d)$ for which $(d-2)(n+1) \leq 2r$, $\phi(i) = (d, m, n, p)$, $c \circ \sigma \in \mathcal{N}_p$, and $c(j)\ G_{m,n}\ c(k)$. By passing to an infinite subset of $I'$, we can assume that the values of $j$, $k$, $n$, and $\sigma$ are independent of $i' \in I'$. Then elements of $I'$ corresponding to the same value of $m$ also correspond to the same value of $i$. As the distance between any two such elements of $I'$ is at most $2r$, it follows that $c \in H_{m,n}$ for infinitely many $m \in \mathbb{N}$, so $c \in H_n$, thus $c \in \mathrm{Hyp}^d(G)$. □

The closure $F$ of the $S$-saturation of $\pi(2^{\mathbb{N}})$ is clearly perfect.

**Lemma 1.5.** *The sequence $0^{\mathbb{Z}}$ is the unique element of $F$ not in $[\pi(2^{\mathbb{N}})]_S$.*

*Proof.* If $(c_n)_{n\in\mathbb{N}}$ is a sequence of elements of $2^{\mathbb{N}}$ and $(k_n)_{n\in\mathbb{N}}$ is a strictly increasing sequence of elements of $\mathbb{N}$, then $\pi(S^{-k_n}(c_n)) \to 0^{\mathbb{Z}}$. Moreover, if $\pi(S^{k_n}(c_n))$ converges to a sequence $c$ other than $0^{\mathbb{Z}}$, then $c(n) = 1$ for a unique $n \in \mathbb{Z}$, thus it is in the $S$-orbit of $\pi(0^{\mathbb{N}})$. □

Lemma 1.5 implies that the union of the supports of finitely many elements of $F$ is contained in a finite union of translates of $\{0\} \cup \bigcup_{i \in I} N_i$. But such sets necessarily have arbitrarily long gaps, so every finite sequence of elements of $F$ is proximal. □

**Proposition 1.6** ($\mathtt{ZF} + \mathtt{DC}$)**.** *Suppose that $d \geq 2$, $X$ and $Y$ are Polish spaces, $G$ is a Borel graph on $X$, $T\colon Y \to Y$ is a Borel automorphism, $B \subseteq Y$ is an uncountable Borel $d$-scrambled set, $\pi\colon X \to Y$ is a Borel homomorphism from $\mathrm{IND}^{\geq 3}_{\mathrm{G}}$ to $\sim\mathrm{LY}_{\mathrm{T}}$, and $[\pi(X)]_T$ is co-countable. Then $G$ has a perfect clique.*

*Proof.* Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)(X)$ is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems

13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a $G$-clique or $G$-independent. But $G$-independence would contradict the fact that $B$ is $d$-scrambled. $\square$

## 2. (<ℕ)-Scrambled sets

We begin this section with the proof of our primary result:

*Proof (of Theorem 1).* By [KV12, Theorem 2.1] and a simple Baire category argument, there is an $F_\sigma$ graph $G$ on $2^{\mathbb{N}}$ with an uncountable clique $Y \subseteq 2^{\mathbb{N}}$ but no perfect clique. By Theorem 2, there is a continuous injective homomorphism $\pi \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ from $(\mathrm{CLQ}^{\geq 3}_{\mathrm{G}}, \mathrm{IND}^{\geq 3}_{\mathrm{G}})$ to $(\mathrm{LY}_{\mathrm{T}}, {\sim}\mathrm{LY}_{\mathrm{T}})$ with $|{\sim}[\pi(2^{\mathbb{N}})]_T| = 1$. Then $\pi(Y)$ is $(<\mathbb{N})$-scrambled, but Proposition 1.6 rules out a perfect 3-scrambled set. $\square$

Let $\mathfrak{c}$ denote the cardinality of the continuum.

**Theorem 2.1** ($\mathtt{ZFC} + \neg\mathtt{CH}$)**.** *It is consistent that there is a homeomorphism $T \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ with a $(<\mathbb{N})$-scrambled set of cardinality $\mathfrak{c}$ but no perfect 3-scrambled set.*

*Proof.* By [She99, Theorem 1.13], we can assume that there is a $F_\sigma$ graph $G$ on $2^{\mathbb{N}}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

**Theorem 2.2** ($\mathtt{ZFC} + \neg\mathtt{CH}$)**.** *Suppose that $d \geq 3$. Then it is consistent that every Borel function on a Polish space with a $d$-scrambled set of cardinality $\aleph_2$ has a $d$-scrambled perfect set.*

*Proof.* By [KS03, Proposition 3.4], we can assume that every Borel $d$-ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But $d$-ary Li–Yorkeness is a Borel hypergraph. $\square$

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